

PROVISIONAL PATENT SPECIFICATION (MASTER)

Title: Autonomous AI-Based Tender Processing and Submission System

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International Patent Classification (IPC)

- G06Q 10/00 – Administration and business processing systems
- G06Q 30/00 – Commerce, quotation generation and procurement systems
- G06N 20/00 – Artificial intelligence and automated decision-making systems

1. FIELD OF THE INVENTION

This invention relates to artificial intelligence systems, automated procurement platforms, and autonomous decision-making systems for tender processing.

2. BACKGROUND OF THE INVENTION

Traditional tendering processes require manual searching, evaluation, pricing, document preparation, and submission. These processes are inefficient, inconsistent, and not scalable. Existing digital systems fail to provide a fully autonomous, end-to-end solution.

3. PROBLEM TO BE SOLVED

There is a need for a system capable of autonomously identifying, evaluating, pricing, generating, and submitting tender responses without human intervention.

4. SUMMARY OF THE INVENTION

The invention provides a fully autonomous tender processing system comprising data ingestion, filtering, scoring, pricing, document generation, and submission modules operating as an integrated pipeline.

5. SYSTEM ARCHITECTURE (FIGURE 1)

- Tender Harvester Module
- Eligibility and Filtering Engine
- Scoring Engine
- Pricing Engine
- Quote Generation Module
- PDF Packaging Module
- Submission Engine
- Automation Controller

These modules operate in a pipeline architecture controlled by an automation layer.

6. PROCESS FLOW (FIGURE 2)

- Retrieve tender data
- Normalize data
- Apply eligibility filtering
- Score opportunities
- Generate pricing
- Create quotation documents
- Convert to PDF
- Submit electronically

7. DECISION ENGINE (FIGURE 3)

The scoring engine evaluates tenders based on weighted parameters including profitability, geographic relevance, compliance, and category. Only qualifying tenders proceed to pricing.

8. TECHNICAL IMPLEMENTATION

The system is implemented using APIs, databases, asynchronous task processing, server-based architecture, and automated document generation libraries. The system operates continuously through scheduled and event-driven execution.

9. VARIATIONS AND IMPLEMENTATIONS

The system may be implemented as cloud-based, distributed, API-driven, or mobile-integrated platforms. Modules may be combined, separated, or modified without departing from the invention.

10. CLAIMS

- A computer-implemented system comprising modules for ingestion, filtering, scoring, pricing, document generation, and submission.
- The system evaluates opportunities based on profitability thresholds.
- The system generates pricing with markup and tax calculations.
- The system produces structured PDF quotation documents.
- The system submits documents electronically.
- The system operates autonomously without human intervention.
- A method comprising retrieving, filtering, scoring, pricing, generating, and submitting tenders.

11. ADVANTAGES

- Reduces manual effort
- Improves efficiency
- Enables scalability
- Increases profitability
- Standardizes tender responses

12. INDUSTRIAL APPLICATION

Applicable in government procurement, private sector sourcing, and automated business systems.

13. ABSTRACT

An AI-driven system that autonomously processes procurement opportunities from identification to submission without human intervention.